

Suyash Soni

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SUMMARY

AI/Backend Engineering student who designs and ships **end-to-end backend systems for AI applications** — REST APIs, LLM integration, and real-time data pipelines using FastAPI, PostgreSQL, and Groq LLM APIs. Delivered production-style projects independently from architecture through deployment, with measurable model and system performance. Currently extending this into infra (Docker, CI/CD) and applied LLM engineering (RAG, agentic workflows).

TECHNICAL SKILLS

Backend & APIs: FastAPI, REST APIs, PostgreSQL, SQLite, JWT Authentication, Node.js

LLM / GenAI Integration: Groq LLaMA 3.1, LLM API Integration, Prompt Engineering, CLIP Visual Search, Huggingface

Computer Vision: YOLO11, YOLO8, YOLO-World, DeepSORT, OpenCV, PyTorch, EfficientNet, DeepLabV3+

Languages: Python, C, C++, Java, JavaScript (basic), HTML (basic)

Data & Analytics: NumPy, Pandas, Matplotlib, SQL, MongoDB, Tableau, Jupyter Notebook

Developer Tools: Git, VS Code, Kaggle, Google Colab, Vercel, Figma (basic)

EXPERIENCE

Full Stack Web Development Intern | [Github link\(fork\)](#)

Jun 2025 – Aug 2025

Techbrill Solutions

Remote

- **Delivered** 50+ reusable UI components adopted across client projects, cutting duplicate development effort for the team
- **Maintained** a consistent 50+ commit delivery pace over a 1-month sprint, demonstrating reliable, self-directed execution in a remote setting
- **Optimized** website performance, directly improving client-facing user experience metrics

PROJECTS

Visual Memory Lane | [Github link](#)

May 2026

FastAPI, PostgreSQL, Groq LLaMA 3.1, YOLO11, DeepSORT, React Native

Personal Project

- **Architected** a **FastAPI** backend with **PostgreSQL**, JWT authentication, and CLIP-based visual search, supporting a **camera-agnostic** AI assistant across up to 4 simultaneous live feeds
- **Built** an LLM reasoning layer with Groq LLaMA 3.1 to convert raw detection events into scene-aware suggestions, interval summaries, and natural-language query answers — reducing manual review of camera history to a single chat query
- **Implemented** real-time object detection and per-ID tracking (**YOLO11 + DeepSORT**) with class persistence across frames, feeding structured events directly into the backend pipeline
- **Shipped** a React Native mobile client giving end-to-end cross-platform access to object history, summaries, and live monitoring — full-stack ownership from model to mobile UI

LedgerApp | **Native Android Billing System** | [Github link](#)

Kotlin, Room Database, MVVM, Jetpack (ViewModel, Navigation)

Personal Project

- **Built** an offline-first transaction management app for Indian retail shopkeepers, prioritizing accessibility for low digital-literacy users
- **Implemented** MVVM architecture with Room (KSP) for local persistence, eliminating dependency on network connectivity for core billing operations
- **Resolved** a Gradle/AGP 9.2.1 + Kotlin toolchain incompatibility by migrating the build system from `kapt` to KSP, unblocking the entire build pipeline

Road Network Segmentation from Satellite Imagery | [Github link](#)

March 2026

Python, PyTorch, DeepLabV3+, EfficientNet-B4, OpenCV

Academic Project

- **Built** a road segmentation system using DeepLabV3+ with an EfficientNet-B4 backbone, trained on 5,634 annotated Sentinel-2 image-mask pairs with mixed-precision GPU training
- **Achieved** F1 **0.5136** and IoU **0.3469** on held-out test data (exceeding target thresholds of $F1 \geq 0.50$ and $IoU \geq 0.38$), with **62.6% recall** via a composite Dice/Focal/BCE loss and ASPP multi-scale convolutions

EDUCATION

Vellore Institute of Technology Bhopal

Bhopal, MP

B.Tech in Computer Science Engineering (AI & ML)

Expected July 2027

- **CGPA:** 8.27/10

ACHIEVEMENTS AND CERTIFICATIONS

Competitive Programming - *LeetCode & CodeChef* | 350+ DSA Problems Solved

Flipkart Grid 6.0 Challenge - *ML Problem Statement* | Score: 89.43718/100, R^2 : 95.0549 / 100

Internship Completion Letter - *TechBrill Solutions* | 3 months

[View Certificate](#)

Applied Machine Learning in Python - *University of Michigan* | 31 Hours

[View Certificate](#)

Cloud Computing (Elite Certificate) - *NPTEL, IIT Kharagpur* | 64% Score, Top Percentile

[View Certificate](#)

Geodata Processing using Python - *IIRS, ISRO* | 7.5 Hours, 100% Completion

[View Certificate](#)